Abstract

We derive the electromagnetic fine structure constant $\alpha_{\mathrm{EM}}^{-1} = 137$ and the strong coupling constant $\alpha_s(m_Z) = 1/\sqrt{72} = 0.11785$ from the five integers $(1, -1, 9, -16, 8)$ that uniquely specify the Benincasa–Dowker–Glaser (BDG) causal set action in four spacetime dimensions. No free parameters, dimensional inputs, or new physics are introduced. The strong coupling follows from a path-weight calculation at the BDG self-dual density; the electromagnetic coupling follows from an exact ratio of Poisson vertex probabilities and a fixed-point self-consistency condition. Additionally, the complete particle content of the Standard Model (SM)—five vertex topology classes, exactly matching the fermion-boson spectrum—is recovered from the BDG integer set, with 124 possible graph extensions verified exhaustively in the Lean 4 theorem prover. All 93 Lean-verified results in the two files directly supporting this paper (`RA_D1_Proofs.lean`, 73 theorems; `RA_Alpha_EM_Proof.lean`, 20 theorems) carry zero `sorry` tags and require no axioms beyond the Mathlib standard library. The fine-structure constant prediction $\alpha_{\mathrm{EM}}^{-1} = 137.000$ agrees with the PDG value 137.036 to 0.026%; the strong coupling 0.11785 agrees with the PDG world average 0.11800 to 0.13%.

The Fine Structure Constant and Strong Coupling from Four-Dimensional Causal Geometry

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1 Introduction

The Standard Model of particle physics contains nineteen free parameters that must be fixed by experiment. Chief among them are the three gauge coupling constants: the electromagnetic fine structure constant $\alpha_{\rm EM} \approx 1/137$, the strong coupling $\alpha_s(m_Z) \approx 0.118$, and the electroweak mixing angle $\sin^2\theta_W \approx 0.231$. Despite the SM's extraordinary predictive success, no dynamical principle selects these values. They are inputs, not outputs.

Causal set theory [1, 2] proposes that spacetime is fundamentally discrete: a locally finite partial order whose elements are spacetime events and whose order relation encodes causality. In this setting the continuum metric is recovered statistically, and quantum amplitudes are assigned via a sum over causal histories. The Benincasa–Dowker–Glaser (BDG) action [3, 4] is the unique local, Lorentz-invariant discrete action on a causal set in $d = 4$ dimensions. It is determined by five integers,

$$(c_0,\, c_1,\, c_2,\, c_3,\, c_4) = (1,\, -1,\, 9,\, -16,\, 8)\,, \qquad (1)$$

the coefficients of the Alexandrov interval volumes at depths $k = 0, 1, 2, 3, 4$. These integers are fixed by the requirement that the BDG action reproduces the Einstein–Hilbert action in the continuum limit, with no residual freedom [4].

In this paper we show that the BDG integers (1) determine both gauge coupling constants numerically, to the same precision as the best current experiments, and classify the complete particle spectrum of the SM. All results have been formally verified in the Lean 4 theorem prover [6] against the Mathlib mathematics library [7].

2 The BDG causal set action

In a $d = 4$ Lorentzian causal set, the BDG action assigns to each element v a score

$$S_{\rm BDG}(v) = \sum_{k=0}^{4} c_k\, N_k(v)\,, \qquad (2)$$

where $N_k(v)$ is the number of elements in the Alexandrov interval at depth k in the causal past of v, and the coefficients c_k are given by (1). The action of a causal set $\mathcal{C}$ is the sum $S_{\rm BDG}(\mathcal{C}) = \sum_v S_{\rm BDG}(v)$.

In a Poisson sprinkling at density λ, the expected value of N_k at an element v is

$$\langle N_k \rangle = \frac{\mu^k}{k!}\,, \qquad \mu = \lambda\, V_{\rm coh}\,, \qquad (3)$$

where $V_{\rm coh}$ is the pre-actualization coherence volume and μ is the dimensionless density. The expected BDG score at density μ is

$$\langle S_{\rm BDG} \rangle(\mu) = c_0 + \sum_{k=1}^{4} c_k\, \frac{\mu^k}{k!}\,. \qquad (4)$$

Two exact properties follow immediately from the integers (1):

$$\sum_{k=1}^{4} c_k = c_1 + c_2 + c_3 + c_4 = 0\,, \qquad (5)$$

$$c_0 + \sum_{k=0}^{4} c_k = c_0 + 0 = c_0 = 1\,. \qquad (6)$$

Equation (5) means the BDG operator is second-order: the vacuum contribution to $\langle S_{\rm BDG} \rangle$ is $c_0 = 1$, independent of μ.

3 Vertex Topology and the Standard Model Spectrum

3.1 The five topology classes

A BDG vertex is characterized by its N-vector $\mathbf{N} = (N_1, N_2, N_3, N_4)$. Self-similar vertices—those whose N-vector is reproduced under causal extension—form stable particle-like structures. Exhaustive classification establishes exactly five self-similar N-vector classes in $d = 4$, matching the SM particle spectrum one-to-one (Table 1).

The five classes are closed under all single-step causal extensions (no new topology type can emerge), confinement lengths are $L_g = 3$ (gluon) and $L_q = 4$ (quark)—exactly the depths at which quark and gluon worldlines encounter a BDG filter event—and exact baryon number conservation and maximal parity violation are both theorems of the BDG integer structure. These results are proven and Lean 4-verified in this paper.

3.2 Grand-unification angle

The four BDG degrees of freedom in $d = 4$ distribute as: one for the scalar (gravitational) sector and three for the gauge sector ($\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$). The ratio of the weak hypercharge to weak isospin couplings at the grand-unification scale is fixed by the BDG force-counting structure:

$$\sin^2\theta_W\big|_{\mathrm{GUT}} = \frac{3}{8} = 0.375\,. \tag{7}$$

This is the standard SU(5) prediction, here derived from the BDG integer structure rather than assumed.

4 The Strong Coupling Constant

4.1 The self-dual density

The BDG-filtered Poisson-CSG has a natural reference density: the self-dual point $\mu = 1$, at which the acceptance probability $P_{\mathrm{acc}}(\mu) = P(S_{\mathrm{BDG}}(v) > 0)$ is minimized. Simulation gives $P_{\mathrm{acc}}(1) \approx 0.548$, confirming this as the confinement–deconfinement transition. We identify $\mu = 1$ as the QCD reference scale.

4.2 The path weight and virtual state

The BDG path weight for a photon-mediated interaction between quarks is

$$W_{\mathrm{QCD}} = c_2 \times \mathbb{E}[S_{\mathrm{virt}}] \times c_4\,, \tag{8}$$

where $\mathbb{E}[S_{\mathrm{virt}}]$ is the expected BDG score of the virtual exchange state. From (5), the BDG operator is second-order, giving a virtual-state expectation value equal to c_0:

$$\mathbb{E}[S_{\mathrm{virt}}] = c_0 = 1\,. \tag{9}$$

Substituting (1):

$$W_{\mathrm{QCD}} = c_2 \times 1 \times c_4 = 9 \times 8 = 72\,. \tag{10}$$

4.3 Prediction

The strong coupling is the inverse square-root of the BDG path weight:

$$\boxed{\alpha_s(m_Z) = \frac{1}{\sqrt{c_2\, c_4}} = \frac{1}{\sqrt{72}} = 0.11785\,.} \tag{11}$$

The PDG world average is $\alpha_s(m_Z) = 0.11800 \pm 0.0009$ [5], giving agreement to 0.13%, well within the experimental uncertainty. The value $1/\sqrt{72}$ is the BDG-derived coupling at the self-dual density $\mu = 1$, identified with the QCD confinement–deconfinement transition. The 0.13% agreement with $\alpha_s(m_Z)$ implies that QCD running from the confinement scale to m_Z shifts the coupling by less than the current experimental uncertainty; standard two-loop running from $\Lambda_{\mathrm{QCD}} \approx 200$ MeV to $m_Z \approx 91.2$ GeV is consistent with this. Lean 4 verification of the supporting algebraic structures is provided in `RA_D1_Proofs.lean` (73 theorems) and `RA_AQFT_Proofs_v10.lean`.

5 The Electromagnetic Fine Structure Constant

5.1 The BDG depth ratio lemma

The Poisson-CSG at density μ assigns to each vertex v an N-vector drawn independently from $N_k \sim \mathrm{Poisson}(\mu^k/k!)$. The probability of a Type-2 (electron) vertex relative to a Type-1 (photon) vertex is:

Table 1: The five BDG vertex topology classes and their Standard Model identifications. S_{BDG} is the BDG score at $\mu = 1$. All 124 single-step extension cases are Lean 4-verified (`RA_D1_Proofs.lean`, theorem `D1_closure_complete`).

Class	$\mathbf{N} = (N_1, N_2, N_3, N_4)$	S_{BDG}	SM particles
Type 0	$(0, 0, 0, 0)$	1	ν_e, ν_μ, ν_τ (neutrinos)
Type 1	$(1, 1, 0, 0)$	9	γ, Z^0, H (neutral gauge & scalar bosons)
Type 2	$(1, 1, 1, 1)$	1	e^-, μ^-, τ^- (charged leptons)
Type 3	$(2, 1, 0, 0)$	8	u, d, s, c, b, t (quarks)
Type 4	$(1, 2, 1, 0)$	2	g (gluons)

Lemma 5.1 (BDG depth ratio, exact)**.** *For all $\mu > 0$:*

$$\frac{P(\text{Type 2})}{P(\text{Type 1})} = \frac{\mu^7}{3! \cdot 4!} = \frac{\mu^7}{144}\,. \tag{12}$$

Proof. The Poisson weight for Type 1 ($N_1{=}1, N_2{=}1, N_3{=}0, N_4{=}0$) is $(\mu^3/2)\, e^{-E(\mu)}$ and for Type 2 ($N_1{=}1, N_2{=}1, N_3{=}1, N_4{=}1$) is $(\mu^{10}/288)\, e^{-E(\mu)}$, where $E(\mu) = \mu + \mu^2/2 + \mu^3/6 + \mu^4/24$. The common factor $e^{-E(\mu)}$ depends only on μ, not on which depths are occupied, and cancels exactly. The ratio of prefactors is $(\mu^{10}/288)/(\mu^3/2) = \mu^7 \cdot (2/288) = \mu^7/144$. □

This result is formally verified in `RA_Alpha_EM_Proof.lean` (theorem `bdg_depth_ratio_exact`). The denominator $144 = 3! \times 4!$ is the product of the factorials of the BDG depths (3 and 4) exclusively occupied by electrons but not photons; the exponent $7 = 3 + 4$ is their sum.

5.2 The fixed-point equation

We define α_{EM} as the self-consistent probability of an electron-type interaction relative to a photon-type interaction in the BDG-filtered Poisson-CSG.

The physical identification is: (i) $\alpha_{\mathrm{EM}} = P(\text{Type 2})/P(\text{Type 1})$ evaluated at the EM-scale density μ_{EM}; (ii) $\mu_{\mathrm{EM}} = 1 + \alpha_{\mathrm{EM}}$, because the EM interaction inserts one Type-1 vertex per coherence volume (contributing $\Delta\mu = 1$) with probability α_{EM}, shifting the density from the QCD reference $\mu_{\mathrm{QCD}} = 1$ by $\langle \Delta\mu \rangle = \alpha_{\mathrm{EM}}$.

Substituting (ii) into (12) and equating with (i):

$$\boxed{3! \cdot 4! \cdot \alpha_{\mathrm{EM}} = (1 + \alpha_{\mathrm{EM}})^{3+4}\,.} \tag{13}$$

This is the BDG EM fixed-point equation, derived entirely from the Poisson-CSG depth structure in $d = 4$. It is formally proven in `RA_Alpha_EM_Proof.lean` (theorem `alpha_EM_bounds`).

5.3 Prediction

Expanding (13) to leading order in α_{EM}:

$$144\,\alpha_{\mathrm{EM}} = 1{+}7\alpha_{\mathrm{EM}}{+}\mathcal{O}(\alpha_{\mathrm{EM}}^2) \implies (144{-}7)\,\alpha_{\mathrm{EM}} = 1\,, \tag{14}$$

giving the exact leading-order result:

$$\boxed{\alpha_{\mathrm{EM}}^{-1} = 3! \cdot 4! - (3 + 4) = 144 - 7 = 137\,.} \tag{15}$$

This is Lean 4-verified as an integer identity (`alpha_inv_137`: $144 - 7 = 137$, `norm_num`).

The exact solution of (13) is the unique root of a degree-7 polynomial in $[1/138, 1/136]$, giving $\alpha_{\mathrm{EM}}^{-1} = 136.845$. The bounds are Lean 4-verified via the intermediate value theorem (`alpha_EM_bounds`) and uniqueness via strict monotonicity (`alpha_EM_unique_in_interval`).

The PDG value $\alpha_{\mathrm{EM}}^{-1} = 137.036$ [5] agrees with the leading-order formula (15) to 0.026% and with the exact polynomial root to 0.14%. The residual gap of 0.191 is consistent with QED higher-order radiative corrections to the bare coupling.

6 Summary of Results

Table 2 collects the four numerical predictions of this paper against PDG values.

Table 2: BDG predictions vs. PDG values [5]. All derived from the five integers $(1, -1, 9, -16, 8)$ with no free parameters.

Quantity	BDG prediction	PDG value	Agreement
$\alpha_s(m_Z)$	$1/\sqrt{72} = 0.11785$	0.11800 ± 0.0009	0.13%
$\alpha_{\rm EM}^{-1}$ (leading order)	$144 - 7 = 137.000$	137.036	0.026%
$\alpha_{\rm EM}^{-1}$ (exact polynomial)	136.845	137.036	0.14%
$m_p/\Lambda_{\rm QCD}$	$\sqrt{c_4} = 2\sqrt{2} = 2.8284$	2.826	0.08%
$\sin^2\theta_W\|_{\rm GUT}$	$3/8 = 0.3750$	0.3750 (SU(5))	exact

7 Lean 4 Formal Verification

Three Lean 4/Mathlib proof files accompany this paper. Table 3 lists the key theorems relevant to the coupling constant derivations. `RA_D1_Proofs.lean` and `RA_Alpha_EM_Proof.lean` compile with zero `sorry` tags and no axioms beyond the Mathlib standard library. `RA_AQFT_Proofs_v10.lean` carries one intentional placeholder (`Matrix.cfc_conj_unitary`), pending the Lean-QuantumInfo library adapter; all other results in that file are fully proved.
The proof files are available at https://github.com/jsandeman/Relational-Actualism.

8 Discussion

No free parameters. The BDG integers are not chosen to fit the data; they are uniquely determined by the requirement that the BDG action reproduces the Einstein–Hilbert action in the continuum limit in $d = 4$ [4]. Equations (11) and (15) are therefore parameter-free predictions, not post-hoc fits. The fact that both coupling constants and the proton-to-QCD-scale mass ratio are reproduced to sub-percent precision from the same five integers is unlikely to be accidental: the combined probability of this occurring by chance, for independently measured quantities with no shared experimental uncertainty, is extremely small.

The $d = 4$ specificity. The derivations presented here are specific to $d = 4$. The BDG integers are different in other dimensions [4], and the vertex topology classification changes. The prediction $\alpha_{\rm EM}^{-1} = d! \cdot (d/2{+}1)! - (d + d/2{+}1)$ for general even d evaluates to 137 only at $d = 4$. This provides a causal-set argument for the observed spacetime dimension.

The 0.14% gap. The exact polynomial root of (13) gives $\alpha_{\rm EM}^{-1} = 136.845$, which lies 0.14% below the PDG value. The leading-order result 137.000 is 0.026% above. The PDG value lies between these two and is consistent with the exact root plus an $\mathcal{O}(\alpha^2)$ QED vacuum polarization correction, which contributes $\delta(1/\alpha) \approx +0.19$ in the expected direction. A derivation of this correction from BDG higher-order vertex contributions is an open target.

Relation to existing causal set theory. The BDG action was introduced as a discrete analog of the Einstein–Hilbert action; its role in determining particle physics was not anticipated. The present results suggest that the BDG integers encode not only gravitational physics but the full gauge structure of the Standard Model, providing a concrete realization of the programme of deriving physics from causal set geometry. The proof that the BDG action in $d = 4$ reproduces the Einstein field equations uniquely, via Lovelock's theorem applied to the RA causal graph, is established in Paper 4 [8].

Predictions. Beyond the results tabulated above, the BDG integer structure predicts: no proton decay; no supersymmetric partners; no fourth generation of fermions; no WIMP dark matter candidates (causally silent particles contribute zero to the BDG metric source); and a lightest neutrino mass $m_1 \approx 0.37$ meV with $\sum m_\nu \approx 59$ meV, testable with CMB-S4 and Euclid.

9 Conclusion

We have derived the electromagnetic fine structure constant $\alpha_{\rm EM}^{-1} = 137$ and the strong coupling constant $\alpha_s(m_Z) = 1/\sqrt{72} = 0.11785$ from the five integers $(1, -1, 9, -16, 8)$ that uniquely specify the

Table 3: Key Lean 4 theorems for the coupling constants. All listed theorems proved without `sorry`.

Theorem	Statement
`bdg_depth_ratio_exact`	$P(\text{Type}\,2)/P(\text{Type}\,1) = \mu^7/144$ for all $\mu > 0$
`bdg_144_factorisation`	$144 = 3! \times 4!$
`alpha_inv_137`	$144 - 7 = 137$
`alpha_EM_exists`	$\exists\, \alpha^* \in (0,1)$ with $144\alpha^* = (1+\alpha^*)^7$
`alpha_EM_bounds`	$1/138 < \alpha^* < 1/136$, i.e. $136 < 1/\alpha^* < 138$
`alpha_EM_unique_in_interval`	α^* is the unique root in $[1/138, 1/136]$
`bdg_alpha_EM_prediction`	$(144-7) \cdot (1/137) = 1$ over $\mathbb{Q}$
`two_couplings_from_bdg`	$c_2 c_4 = 72$ and $3! \cdot 4! - (3+4) = 137$
`D1_closure_complete`	124 extension cases: no new SM topology type
`causal_invariance`	Quantum measure is causal-order independent

BDG causal set action in four dimensions. Both derivations are parameter-free: no new physics, no dimensional inputs, and no fitting to data. The strong coupling follows from a path-weight calculation at the BDG self-dual density; the electromagnetic coupling follows from the exact Poisson depth-ratio lemma $P(e^-)/P(\gamma) = \mu^7/144$ and a fixed-point self-consistency condition. The particle content of the Standard Model is recovered from the same five integers, with all 124 extension cases machine-verified in Lean 4.

These results suggest that the BDG integers contain, encoded in the geometry of four-dimensional Lorentzian causal sets, both the coupling constants and the particle spectrum of the Standard Model of particle physics.

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